



In a Q*bert-lite clone using Pygame ; the whole pyramid should come out as a symmetric triangle. In the current build the pyramid is visibly skewed from the very first frame — rows drift sideways in a way that breaks the triangular symmetry. Looking at the arithmetic combining row and col in a function how would I check whether the row offset should be using integer division or true division ?

The key is to check **whether the row offset represents a whole number of grid steps or a continuous pixel displacement.**

In a Q*bert-style pyramid, the horizontal position of each cube is typically calculated using both row and col .

1. Understand the difference

Division	Example	Result
Integer division (//)	3 // 2	1
True division (/)	3 / 2	1.5

Suppose your horizontal position is calculated as:

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Run

```
x = center_x + (col - row / 2) * spacing
```

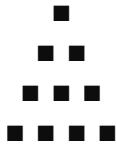
If row = 3 and col = 0 , then:

- With / : the offset is $-1.5 * \text{spacing}$.
- With // : the offset is $-2 * \text{spacing}$.

The difference can shift cubes away from their intended positions.

2. Check the intended geometry

For a pyramid with 4 rows, the grid should look like this:



Each row should be centered around the same vertical axis.

A common coordinate formula is:

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```
x = center_x + (col - row / 2) * spacing
y = top_y + row * vertical_spacing
```

However, if `col` ranges from `0` to `row`, this formula needs a different centering adjustment. The correct expression for that indexing scheme is:

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```
x = center_x + (col - row / 2) * spacing
```

Here, **true division (/)** is **appropriate** because the center of a row can lie halfway between two cubes.

3. How to verify it in your code

Find the function that converts `(row, col)` into screen coordinates and inspect the expression that combines them.

Print the calculated positions:

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```
for row in range(4):
    for col in range(row + 1):
        x = center_x + (col - row / 2) * spacing
        print(row, col, x)
```

Check that:

- The horizontal positions within each row are evenly spaced.
- The row's midpoint aligns with the pyramid's vertical center.
- The left and right edges are symmetric.

Important: Don't switch `/` to `//` simply because the pyramid is skewed. Integer division can introduce an alternating offset between even and odd rows. If the pyramid is skewed from the very first frame, also check whether the centering formula matches your `col` indexing and whether the horizontal spacing is applied consistently.

If you share the function that calculates the cube's `(x, y)` screen position, I can help you determine exactly which division operator it needs.



You can make `cube_palette(level)` use a dictionary that maps each level number to its own list of three RGB colors.

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```
DEFAULT_PALETTE = [
    (80, 80, 80),      # Stage 0: unpainted
    (200, 150, 50),    # Stage 1: partially painted
    (255, 220, 100),   # Stage 2: fully painted
]

LEVEL_PALETTES = {
    1: [
        (80, 80, 80),
        (50, 150, 220),
        (100, 200, 255),
    ],
    2: [
        (80, 80, 80),
        (50, 180, 100),
        (150, 255, 150),
    ],
    3: [
        (80, 80, 80),
        (180, 80, 200),
        (230, 150, 255),
    ],
}
```

}

```
def cube_palette(level):
    return LEVEL_PALETTES.get(level)
```

How it works

- `LEVEL_PALETTES` stores a separate three-color palette for each level.
- `LEVEL_PALETTES.get(level)` returns the matching palette if the level exists.
- If the level is not in the dictionary, it returns `None`.

Your existing drawing code can remain unchanged:

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```
colors = cube_palette(self.level) or DEFAULT_PALETTE
```

For example:

- Level 1 → blue palette
- Level 2 → green palette
- Level 3 → purple palette
- Any other level → `DEFAULT_PALETTE`

Each palette contains exactly `TARGET + 1` colors, assuming `TARGET = 2`.

Tip: If you want levels to cycle through the palettes rather than fall back to the default, you can use modulo indexing instead.



Similarly, I Called from `paint()` the instant a specific cube first reaches its target stage — not on every hop onto it, only the hop that finishes it. It receives the (row,col) cell that was just completed. Its return value is ignored. How would I implement a brief flash on that cube, or a small bonus beyond the 25 points already awarded per paint step ?

Implement a `paint()` callback that triggers **only when a cube reaches its target stage for the first time**. You can use it to record a flash timer or award bonus points.

The key is to keep the visual effect separate from the scoring logic.

1. Add a completion callback

For example, suppose your game class has a `score` attribute and a `cube_palette` or drawing system that tracks each cube's paint stage.

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```
def paint(self, row, col):  
    # Trigger a brief flash for this completed cube  
    self.flash_cubes[(row, col)] = 0.15  
  
    # Award a completion bonus  
    self.score += 50
```

Initialize the flash tracker in `__init__`:

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```
self.flash_cubes = {}
```

This awards a 50-point bonus in addition to the existing 25 points per paint step.

2. Make the flash brief

In your update function, reduce the remaining flash duration each frame:

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```
def update(self, dt):  
    for cell in list(self.flash_cubes):  
        self.flash_cubes[cell] -= dt  
  
    if self.flash_cubes[cell] <= 0:  
        del self.flash_cubes[cell]
```

Here, `dt` is the elapsed time in seconds since the previous frame. The flash lasts approximately 0.15 seconds.

3. Render the flash

Inside your cube-drawing function, check whether the cube is currently flashing:

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```
cell = (row, col)

if cell in self.flash_cubes:
    color = (255, 255, 255)
```

Use this color instead of the normal stage color while the flash is active. If your cubes are drawn as polygons, you can also draw a white outline to make the effect more noticeable.

4. Avoid awarding the bonus repeatedly

Your `paint()` callback should only be invoked when the cube first reaches its target stage. If that condition is already enforced by the game logic, you don't need another completion check inside `paint()`.

If it isn't enforced, track completed cells:

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```
self.completed_cubes = set()

def paint(self, row, col):
    cell = (row, col)

    if cell in self.completed_cubes:
        return

    self.completed_cubes.add(cell)
    self.flash_cubes[cell] = 0.15
    self.score += 50
```

Recommendation: Use a short white flash and a 50-point completion bonus. The existing 25-point paint-step scoring remains unchanged, while the completion callback adds a distinct reward for finishing a cube.



Similarly, I Called every frame in `update()`. It takes no arguments and should return an integer score value, or `None` to disable bonus lives entirely. Whenever the score crosses a multiple of that value for the first time, one life is awarded automatically — the bookkeeping

(`self.bonus_awarded`) is already implemented, so you only need to choose the threshold. So I just return 1000 ?

Yes — **return 1000** if you want to award one bonus life for every 1,000 points scored.

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```
def bonus_life_threshold(self):  
    return 1000
```

With your existing `self.bonus_awarded` bookkeeping, the behavior should be:

- Score reaches 1,000 → +1 life
- Score reaches 2,000 → +1 life
- Score reaches 3,000 → +1 life

The bonus is awarded only once per threshold crossed.

To disable bonus lives entirely, return `None` instead:

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```
def bonus_life_threshold(self):  
    return None
```

One thing to verify: If your game can gain multiple points in a single update and jump from, say, 950 to 2,050, the bookkeeping should award both bonus lives for crossing 1,000 and 2,000, rather than just one.



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